

A DYADIC GRAPH CONSTRUCTION FOR DISTINCT SUBSET SUMS

ABSTRACT. We construct, for every $n \geq 2$, a set of n positive integers with distinct subset sums and maximum less than $2^{n+1}/\log_2 n$. The construction uses a dyadic directed graph with a small determinant and an integer lattice excluding balanced relations. Binary padding converts the associated modular labels into positive integers.

1. INTRODUCTION

A finite set A of positive integers is *sum-distinct* if the map $S \mapsto \sum_{a \in S} a$ is injective on the subsets of A , including the empty subset. For positive integers n, N , define

$$f(n) = \min_{\substack{|A|=n \\ A \text{ sum-distinct}}} \max A, \quad F(N) = \max_{\substack{A \subseteq \{1, \dots, N\} \\ A \text{ sum-distinct}}} |A|.$$

All logarithms are to base 2. Counting the 2^n subset sums in $[0, n \max A]$ gives $f(n) \geq (2^n - 1)/n$, whereas the powers of two give $f(n) \leq 2^{n-1}$.

1.1. Background and history. The problem appears in Erdős's 1956 survey, where he asks whether $F(N) = \log N + O(1)$ and reports a second-moment argument with Moser giving $f(n) \gg 2^n/\sqrt{n}$ [4]. The proposed estimate for F is equivalent to a lower bound $f(n) \geq c2^n$ for some absolute constant $c > 0$. On the construction side, Conway and Guy introduced a recursive sequence giving $f(n) < 2^{n-2}$ for sufficiently large n ; Bohman proved that every set in this sequence is sum-distinct [1]. Bohman's 1998 paper develops a broader construction giving $f(n) < 0.22002 \cdot 2^n$ for sufficiently large n [2].

Dubroff, Fox and Xu [3] proved the lower bound

$$f(n) \geq \left(\sqrt{\frac{2}{\pi}} - o(1) \right) \frac{2^n}{\sqrt{n}},$$

matching an earlier unpublished result of Elkies and Gleason. They give proofs using normal approximation and an isoperimetric inequality.

1.2. Main result. We prove the following quantitative improvement to the upper bound.

Theorem 1.1 (Quantitative bound). *For every integer $n \geq 2$,*

$$f(n) < \frac{2^{n+1}}{\log n}.$$

Consequently $f(n)/2^n \rightarrow 0$, so no positive absolute lower constant is possible. For every integer $N \geq 2^{16}$,

$$F(N) \geq \lfloor \log N + \log \log \log N - 1 \rfloor.$$

1.3. Proof overview. We first seek r residues modulo q whose equal-cardinality subset sums are distinct, with q substantially smaller than 2^r . We allow collisions between subsets of different sizes: requiring all 2^r subset sums to be distinct modulo q would force $q \geq 2^r$ by the pigeonhole principle. For integers $H \geq 1$, Proposition 2.1 supplies such sets with $r \asymp 2^H$ and $q/2^r = O(1/H)$.

Section 2 explains why this weaker property suffices. Choose representatives $0 \leq c'_j < q$ and an integer $t \geq 0$. Add the offset $b = (2^t + r)q$ to each representative, and adjoin the t binary multiples $q, 2q, \dots, 2^{t-1}q$. In an equality of integer subset sums, the offset is too large to be cancelled unless both sides contain the same number of shifted labels. Reduction modulo q then determines those

labels, and binary uniqueness determines the remaining elements. The resulting set A has $r + t$ elements and

$$\frac{\max A}{2^{r+t}} < \left(1 + \frac{r+1}{2^t}\right) \frac{q}{2^r}.$$

Taking 2^t of order r^2 makes the prefactor tend to one, while $r + t = r + O(\log r)$. Thus the saving $1/H$ in the modulus becomes a saving of order $1/\log n$ in the maximum. Doubling the set and adjoining 1 adds one element without changing the normalized maximum, allowing every cardinality.

Section 3 constructs the modular sets from a directed graph on r vertices, joining 2^H ordered stages to a tree of dyadic intervals. Every cycle passes through an entry vertex a , and every vertex has two outgoing edges except a root marker d , which has one. If E is the adjacency matrix, put $M = 2\text{Id} - E$ and $q = \det M$. Start a walk at a , choosing each outgoing edge with probability $1/2$ and interpreting the missing probability at d as death. Stop at death or the first return to a . Then $q/2^r$ is the death probability. The tree lets the walk jump over dyadic blocks. Let u_h be the expected number of stages visited in a block of size 2^h , entered at its left endpoint, divided by 2^h . Splitting blocks into halves gives

$$u_0 = 1, \quad u_h = u_{h-1} - \frac{3}{16}u_{h-1}^2, \quad \frac{q}{2^r} = \frac{3}{8}u_H.$$

The reciprocal $1/u_h$ grows linearly with h , giving the bound of order $1/H$.

The same graph rules out balanced ternary vectors in the row lattice $L = \mathbb{Z}^r M$, where a vector is *balanced* if its coordinates sum to zero. Suppose $p = zM$ has coordinates in $\{-1, 0, 1\}$ and is balanced. The row sums force $z_d = 0$. Each positive coefficient of z requires a positive predecessor, and similarly for negative coefficients. Each nonempty sign class therefore contains a cycle; since every cycle contains a , both signs cannot occur. After reversing signs, a nonzero z would contain a positive cycle. Its transitions supply positive dyadic markers covering the whole interval. Two positive child markers force their parent positive, so the covering forces $z_d > 0$, a contradiction.

It remains to turn the lattice into a single modulus. The determinant is odd, and the principal minor at a is 2^{r-1} . An adjugate column therefore defines a surjection $\mathbb{Z}^r \rightarrow \mathbb{Z}/q\mathbb{Z}$ with kernel exactly L . The images of the coordinate vectors are the desired labels: a balanced modular relation would lie in L , where the preceding argument excludes it.

2. FROM MODULAR SETS TO DISTINCT SUBSET SUMS

Proposition 2.1 (Modular set construction). *For integers $H \geq 1$, one can choose positive odd integers q_H and sets $C_H \subseteq \{0, \dots, q_H - 1\}$ of size $r_H = 2^{H+2} - 1$ such that, for any $S, T \subseteq C_H$,*

$$|S| = |T|, \quad \sum_{c \in S} c \equiv \sum_{c \in T} c \pmod{q_H} \implies S = T. \quad (2.1)$$

Moreover,

$$0 < \frac{q_H}{2^{r_H}} \leq \frac{2}{H + 16/3}, \quad \frac{q_H}{2^{r_H}} \sim \frac{2}{H} \quad (H \rightarrow \infty). \quad (2.2)$$

We deduce Theorem 1.1 from Proposition 2.1 using the following lemmas; the proof of the proposition is deferred to Section 3.

Lemma 2.2 (Binary padding). *Let $q \geq 1$ be an integer, and let $C \subseteq \{0, \dots, q - 1\}$ have size $r \geq 1$ and distinct equal-cardinality subset sums modulo q . For an integer $t \geq 0$, set $b = (2^t + r)q$ and*

$$A = \{2^i q : 0 \leq i < t\} \cup \{b + c : c \in C\}.$$

Then A is sum-distinct, consists of $r + t$ positive integers, and $\max A < (2^t + r + 1)q$.

Proof. Write $C = \{c'_1, \dots, c'_r\}$. The shifted elements are distinct, positive, and larger than every binary element, giving the cardinality and maximum. A collision of subset sums would give

$$\sum_{i=0}^{t-1} \alpha_i 2^i q + b \sum_j \beta_j + \sum_j \beta_j c'_j = 0, \quad \alpha_i, \beta_j \in \{-1, 0, 1\}. \quad (2.3)$$

The first and third terms together have absolute value at most $(2^t - 1)q + r(q - 1) < b$. Since $\sum_j \beta_j$ is an integer, (2.3) forces this sum to be zero. Reduction modulo q shows that the positive and negative coordinates of β select equal-cardinality subsets of C with the same sum modulo q . By hypothesis these subsets coincide; as they are disjoint, both are empty. Thus $\beta = 0$. Then $\alpha = 0$, because the largest nonzero signed power of two exceeds the sum of all smaller powers. $\square$

Lemma 2.3 (Extension by parity). *If A is a nonempty sum-distinct set of positive integers, then $A^+ = \{2a : a \in A\} \cup \{1\}$ is sum-distinct, $|A^+| = |A| + 1$, and $\max A^+ = 2 \max A$.*

Proof. Parity determines whether a subset contains 1; after removing it and dividing by 2, sum-distinctness of A determines the remaining subset. The other assertions follow because all doubled elements are distinct and at least 2. $\square$

Proof of Theorem 1.1. For each integer $H \geq 1$, take q_H, C_H from Proposition 2.1, put $r = r_H = 2^{H+2} - 1$ and $q = q_H$, and enumerate C_H as $c'_1, \dots, c'_r$. Apply Lemma 2.2 with

$$t = \lceil \log(r^2) \rceil = 2H + 4, \quad n_H = r + t = 2^{H+2} + 2H + 3. \quad (2.4)$$

Call the resulting set A_H and its maximum N_H . Since $2^t = (r+1)^2$, Proposition 2.1 and Lemma 2.2 give

$$\frac{N_H}{2^{n_H}} < \left(1 + \frac{r+1}{2^t}\right) \frac{q}{2^r} \leq \frac{2(1+2^{-H-2})}{H+16/3}. \quad (2.5)$$

For $n \geq 13 = n_1$, choose the largest H with $n_H \leq n$ and apply Lemma 2.3 exactly $n - n_H$ times; the ratio of the maximum to 2^n is unchanged. For $H \geq 1$,

$$n < n_{H+1} < 2^{H+4}, \quad (H+4)(1+2^{-H-2}) \leq H+16/3,$$

where the second inequality follows from $(H+4)2^{-H-2} \leq 5/8 < 4/3$. Thus (2.5) gives $f(n)/2^n < 2/(H+4) < 2/\log n$. For $2 \leq n < 13$, use the binary set, since $\log n < 4$. This proves the first bound and the claimed limit.

For the inverse bound, let $N \geq 2^{16}$, $L_0 = \log N$, $T = \log \log L_0$, and $n = \lfloor L_0 + T - 1 \rfloor$. Then $T \geq 2$, $n \geq L_0$, and

$$2^{n+1} \leq 2^{L_0+T} = N \log L_0 \leq N \log n.$$

The first bound gives $f(n) < N$, so $F(N) \geq n$, as required. $\square$

Remark 2.4 (Asymptotics of the constructed sets). The sets A_H in the preceding proof satisfy $N_H/2^{n_H} \sim 2/H$: the ratio of $N_H/2^{n_H}$ to $q/2^r$ lies between 1 and $1+2^{-H-2}$, and Proposition 2.1 gives $q/2^r \sim 2/H$. Since $\log n_H = H + 2 + o(1)$, their maxima are $(2+o(1))2^{n_H}/\log n_H$. The maximal-height parity extensions chosen above have maximum $(2+o(1))2^n/\log n$, since $H = \log n + O(1)$; this describes the construction, not a matching lower bound for f . For $N \geq 2^{16}$, the uniform inverse bound gives $F(N) > \log N + \log \log \log N - 2$. Along the original maxima it sharpens to $F(N_H) \geq n_H = \log N_H + \log \log \log N_H - 1 + o(1)$, because both $\log \log \log N_H$ and $n_H - \log N_H + 1$ equal $\log H + o(1)$.

3. CONSTRUCTION OF THE MODULAR SET

It remains to prove Proposition 2.1. We construct the matrix M from a dyadic graph, estimate its determinant, exclude balanced ternary vectors from its row lattice, and identify the resulting quotient with a cyclic group.

3.1. The dyadic graph. Throughout Section 3, fix an integer $H \geq 1$, and put $m = 2^H$ and $r = 4m - 1$. Let $\mathcal{D}$ consist of the intervals $[t2^h, (t+1)2^h)$ for integers $0 \leq h \leq H$ and $0 \leq t < m/2^h$. Write $e(I)$ for the right endpoint. For $I \neq [0, m)$, let $\text{par}(I)$ be the dyadic interval of twice its length containing it. The vertices are $V = \{x_i, y_i : 0 \leq i < m\} \cup \{w_I : I \in \mathcal{D}\}$. Set $a = x_0$, $d = w_{[0, m)}$, and use $x_m = x_0$ as an alias, not a new vertex. The edges are

$$\begin{aligned} x_i &\longrightarrow y_i, \quad w_{[i, i+1)}, & y_i &\longrightarrow x_{i+1}, \quad w_{[i, i+1)} \quad (0 \leq i < m), \\ w_I &\longrightarrow x_{e(I)}, \quad w_{\text{par}(I)} \quad (I \neq [0, m)), & d &\longrightarrow a. \end{aligned} \quad (3.1)$$

Figure 1 shows the construction for $H = 2$. Let E be the adjacency matrix with outgoing edges in rows, and define $M = 2\text{Id}_r - E$, $q = \det M$, and $\bar{a} = V \setminus \{a\}$.

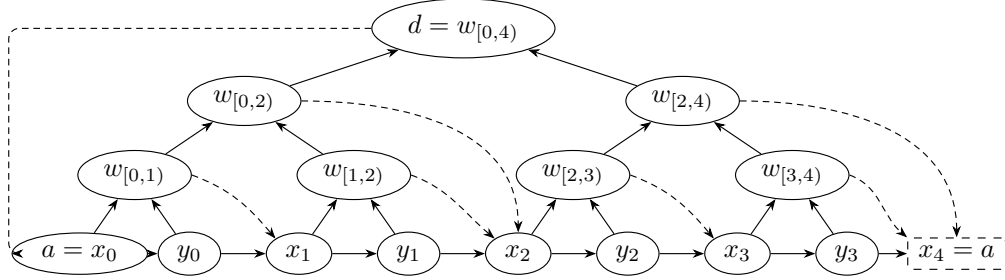


FIGURE 1. The graph for $H = 2$ ($m = 4$, $r = 15$). Solid arrows are stage and parent edges; dashed arrows are marker exits. The right-hand $x_4 = a$ repeats x_0 , not a new vertex.

Lemma 3.1 (Graph structure). *The graph has r vertices and is acyclic after deletion of a . Every row of M has sum zero except row d , which has sum one. Moreover*

$$\det M_{\bar{a}, \bar{a}} = 2^{r-1}. \quad (3.2)$$

Proof. There are $2m$ stage vertices and $2m - 1$ markers. After deleting a , order markers by right endpoint, shorter first at a tie, and place x_i, y_i after markers ending at i and before those ending at $i + 1$. Every edge points forward, so $E_{\bar{a}, \bar{a}}$ is strictly upper triangular. This proves (3.2); the row sums follow from (3.1). $\square$

3.2. The determinant estimate. The next proposition gives the quantitative modulus bound in Proposition 2.1.

Proposition 3.2 (A small positive determinant). *Define $u_0 = 1$ and $u_h = u_{h-1} - (3/16)u_{h-1}^2$ for $h \geq 1$. Then*

$$0 < \frac{q}{2^r} = \frac{3}{8}u_H \leq \frac{2}{H + 16/3}, \quad 0 < u_H \leq \frac{1}{1 + 3H/16}. \quad (3.3)$$

As $H \rightarrow \infty$, one also has $u_H \sim 16/(3H)$ and $q/2^r \sim 2/H$.

Proof. Put $B = E/2$. Interpret B as a transition rule that chooses each outgoing edge with probability $1/2$, with the missing probability $1/2$ at d meaning death. Start at a and stop at the first return to a or at death. By Lemma 3.1, the submatrix $B_{\bar{a}, \bar{a}}$ is nilpotent. Hence $\det(\text{Id} - B_{\bar{a}, \bar{a}}) = 1$ and its inverse is a finite geometric series. The block determinant formula, using $B_{a, a} = 0$, yields

$$\frac{q}{2^r} = 1 - R, \quad R = B_{a, \bar{a}}(\text{Id} - B_{\bar{a}, \bar{a}})^{-1}B_{\bar{a}, a}. \quad (3.4)$$

The power expansion identifies R with the return probability. Every excursion is finite, and death can occur only at d , where return and death are equally likely. It follows that

$$\frac{q}{2^r} = \frac{1}{2}\mathbb{P}_a(\text{hit } d \text{ before returning to } a) > 0. \quad (3.5)$$

The inequality is strict because the path from a to $w_{[0,1)}$ and then through its successive parents reaches d with positive probability.

Record the indices of visits to x_i until a hit to d or a return. These indices increase strictly; a return is an exit at index m , not another visit to index zero. At a visited stage, the walk takes $x_i \rightarrow y_i \rightarrow x_{i+1}$ with probability $1/4$ and reaches the leaf marker with probability $3/4$. At each proper marker it exits to the right endpoint or ascends to the parent, each with probability $1/2$. Thus the height k of the next transition has the following conditional law, independent of the past:

$$\mathbb{P}(k=0) = \frac{5}{8}, \quad \mathbb{P}(k=j) = \frac{3}{4} 2^{-j-1} \quad (1 \leq j < H), \quad \mathbb{P}(k=H) = \alpha_H = \frac{3}{4} 2^{-H}. \quad (3.6)$$

For $k < H$, the next index is $e_k(i) = (\lfloor i/2^k \rfloor + 1)2^k$; height H means a hit to d . In particular $s_h := \mathbb{P}(k \geq h) = (3/4)2^{-h}$ for $1 \leq h \leq H$.

For $0 \leq h \leq H$, let b_h be the expected number of stage visits in a dyadic block of size 2^h entered at its left endpoint, before leaving that block. This expectation depends only on h : heights below h have the same relative endpoints in every aligned block, and all larger heights leave it. We claim

$$b_0 = 1, \quad b_h = 2b_{h-1} - s_h b_{h-1}^2 \quad (1 \leq h \leq H). \quad (3.7)$$

Indeed, the expected visits in the left half are b_{h-1} . A height at least h at any of them skips the right half. These skip events are disjoint, since each exits the block, so their probability is $s_h b_{h-1}$. Otherwise the first index outside the left half is the midpoint, since every smaller-height endpoint is at most the midpoint. Conditional on entering the right half, its expected visits are again b_{h-1} by the transition law. This gives $b_h = b_{h-1} + (1 - s_h b_{h-1})b_{h-1}$, proving (3.7).

Consequently $b_h/2^h$ satisfies the defining recurrence for u_h . On the full block, a hit to d has conditional probability α_H at each visited stage. These events are disjoint, so the hit probability is $\alpha_H b_H$. Equation (3.5) now gives $q/2^r = (3/8)u_H$. Induction in the recurrence gives $0 < u_h \leq 1$, and

$$\frac{1}{u_h} - \frac{1}{u_{h-1}} = \frac{3/16}{1 - (3/16)u_{h-1}} \geq \frac{3}{16}. \quad (3.8)$$

Summing proves (3.3). It also gives $u_H \rightarrow 0$, so the reciprocal increments in (3.8) tend to $3/16$. Averaging the increments yields $(1/u_H)/H \rightarrow 3/16$, which proves both asymptotic equivalences. $\square$

3.3. Excluding balanced lattice relations. The determinant estimate controls the size of the modulus. We next prove the property that will separate equal-cardinality subset sums. Write $L = \mathbb{Z}^r M$ for the integer row lattice of M .

Lemma 3.3 (Balanced ternary avoidance). *If $p \in L$, $p_j \in \{-1, 0, 1\}$ for every $j \in V$, and $\sum_j p_j = 0$, then $p = 0$.*

Proof. Write $p = zM$ with $z \in \mathbb{Z}^r$. By Lemma 3.1, $z_d = \sum_j p_j = 0$. Every vertex with $z_j > 0$ has a positive predecessor, since otherwise $p_j = 2z_j - \sum_{i \rightarrow j} z_i \geq 2$. The analogous assertion holds for negative vertices. Following predecessors in a nonempty sign class produces a directed cycle, and every cycle contains a by Lemma 3.1. Thus both sign classes cannot occur. If $z \neq 0$, reverse signs if necessary so that $z \geq 0$, and choose a simple cycle C of positive vertices. It contains a and avoids d .

Two distinct positive predecessors of j force $z_j > 0$, because

$$2z_j = p_j + \sum_{i \rightarrow j} z_i \geq -1 + 2 = 1. \quad (3.9)$$

Follow C from $x_0 = a$. Each transition between consecutive stages x_i, x_j has $j > i$, with $j = m$ denoting the return to a . In a direct transition $x_i \rightarrow y_i \rightarrow x_{i+1}$, the two positive predecessors force $w_{[i, i+1)}$ positive. Otherwise the transition uses a leaf marker and some ancestors. Its last marker w_J is positive, and its interval contains $[i, j)$, since $i \in J$ and $j = e(J)$. The resulting positive intervals therefore cover all leaves of $[0, m)$.

If a dyadic node K is covered by selected positive intervals contained in K , then either K is selected or those intervals lie in its two children and cover both. Induction on the height, using (3.9) for the two child markers, forces w_K positive. Applying this at the root contradicts $z_d = 0$. Thus $z = 0$ and $p = 0$. $\square$

3.4. Passing to a cyclic modulus. We now identify the lattice L as the kernel of one congruence modulo q . This is the step that produces the modular set.

Proposition 3.4 (Modular labels). *The integer q is odd. There exist distinct residues $c'_j \in \{0, \dots, q-1\}$, indexed by $j \in V$, such that*

$$\beta \in \{-1, 0, 1\}^V, \quad \sum_j \beta_j = 0, \quad \sum_j \beta_j c'_j \equiv 0 \pmod{q} \implies \beta = 0. \quad (3.10)$$

Proof. First work over $\mathbb{F}_2$. Suppose $Ez = 0$ for a column z , and write $v_i = z_{x_i}$, $v_m = v_0$, and $z_I = z_{w_I}$. The d -row gives $v_0 = 0$; the stage and proper marker rows give

$$z_{y_i} = z_{[i, i+1)} = v_{i+1}, \quad z_{\text{par}(I)} = v_{e(I)}.$$

For a non-leaf interval $P = [l, e)$ with midpoint b , its two child rows imply $z_P = v_b = v_e$. Starting with $v_m = 0$ and splitting intervals recursively forces all v_i to vanish, then all marker and y_i coordinates vanish by the displayed equalities. Hence E is invertible over $\mathbb{F}_2$. Since $M \equiv E \pmod{2}$, q is odd.

Let e_a be the standard coordinate column and set $c = \text{adj}(M)e_a$. Then $Mc = qe_a$ and $c_a = 2^{r-1}$ by Lemma 3.1. Therefore

$$\phi : \mathbb{Z}^r \longrightarrow \mathbb{Z}/q\mathbb{Z}, \quad \phi(p) = pc \pmod{q}$$

annihilates L and is surjective, since $\gcd(c_a, q) = 1$. The index $[\mathbb{Z}^r : L]$ is $|\det M| = q$, whereas $\ker \phi$ also has index q . Here the lattice-index formula follows from the fundamental parallelepiped of the rows of M : it has volume $|\det M|$, and every translate of $\mathbb{Z}^r$ has one point per coset modulo L in that half-open domain. Integrating this count over $[0, 1)^r$ gives the index. Finiteness follows already from $q\mathbb{Z}^r \subseteq L$, by the adjugate. Thus $L = \ker \phi$.

Take c'_j to be the least nonnegative residue of c_j . A vector satisfying the hypotheses of (3.10) lies in $\ker \phi = L$, and is zero by Lemma 3.3. Taking $\beta = e_i - e_j$ also proves pairwise distinctness. $\square$

Proof of Proposition 2.1. For each $H \geq 1$, use the graph above and put $q_H = \det M$. Lemma 3.1 gives $r_H = 4 \cdot 2^H - 1 = 2^{H+2} - 1$ vertices. Proposition 3.4 gives an odd q_H and r_H distinct representatives $c'_j \in \{0, \dots, q_H - 1\}$; let C_H be their set. Two equal-cardinality subsets of C_H with the same sum modulo q_H give, by subtraction, a balanced ternary vector satisfying (3.10). That vector is zero, so the subsets coincide. This proves (2.1). Positivity, the finite bound, and the asymptotic equivalence in (2.2) all follow from Proposition 3.2. $\square$

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